Design Study of scFlowVis: Supporting Single-Cell RNA Sequencing Analysis through Visual Analytics

Supplementary Material II: Expert Interview Agenda and Prompts

Yiwen Xing, Stanley Owomero, Sophia Tsoka, Rita Borgo, and Alfie Abdul-Rahman

Part-0: Preliminary Preparation (10 mins)

- Explanation of the goal and setup of the experiment
- Notification of observation and videotaping
- Obtain verbal consent about recording from the participants
- Collect the participants' background information
- Present the tool via a screen-share and demonstrate what everything is:
    - Show how the **Data Structure Viewing Panel (DSV)** works
    - Show how the **Flowchart Visual Programming Panel (FVP)** works
    - Show how the **Visualization Method Recommendation Panel (VMR)** works
    - Show how the **Visualization Results Display Panel (VRD)** works

Part-1: One-to-one User Testing(30 mins)

Protocol:

In this phase, we invite participants to actively explore the tool by engaging in a set of tasks aligned with key stages of the scRNA-seq analysis workflow. While the tasks are presented in a logical order reflecting typical pipeline progression, participants are encouraged to navigate them iteratively—moving back and forth as needed—to better reflect real-world, non-linear analysis practices.

For participants who are unable to join on-site, we facilitate their involvement through online sessions, employing remote-operated testing methods.

During the test, we encourage participant to verbalize their thoughts and processes by thinking aloud while they navigate the tool. Additionally, we request that they share their screens, enabling us to observe their interactions with the software in real-time.

Objectives:

Verify that the tool operates as intended, focusing particularly on the performance and usability of the four distinct panels.

Ascertain the tool's effectiveness in fulfilling the specific needs of the domain it is designed to serve.

Activities:

**Note: While tasks follow a pipeline-inspired structure, participants are encouraged to navigate them flexibly and iteratively—revisiting earlier stages, making comparisons, and testing alternatives—to reflect the exploratory nature of scRNA-seq data analysis.*

Task 1: Data Preprocessing with the pbmc3k Dataset

Objective: Participants will begin by applying data preprocessing techniques to the PBMC3K datasets utilizing the tool.

Instructions: Engage with the tool's array of preprocessing functionalities, aiming to explore and apply as many relevant features as possible. This includes, but is not limited to, normalization, filtering, detection of highly variable genes, calculating QC metrics, and computing neighbors. Participants are also encouraged to explore the relevant visualization-producing functionalities supported by the tool.

Task 2: Dimensionality Reduction and Clustering

Objective: Building upon the preprocessed data, participants will then perform dimensionality reduction and clustering.

Instructions: Utilize the results from the preprocessing task to conduct dimensionality reduction and clustering. Experiment with different methods and parameters to shape the data and resulting visualizations according to your analytical needs. This task encourages creative exploration, aiming to identify significant clusters or trends within the dataset. Participants are encouraged to utilize the flowchart panel and grid display for experimenting with various parameter settings and for juxtaposing distinct visualization outcomes, enhancing both understanding and analysis.

Task 3: In-Depth Analysis of Processed Data

Objective: For the final task, participants are invited to delve deeper into the analysis of the data obtained from the previous tasks. This involves choosing between two advanced analytical approaches – *Differential Expression Analysis* or *Trajectory Inference*.

Instructions: Based on the results of dimensionality reduction and clustering, select either Differential Expression Analysis to identify genes that are significantly varied across conditions or cell types, or Trajectory Inference to infer cellular developmental trajectories. Using the relevant functionalities in the tool to conduct a thorough analysis using the chosen method.

Part 2: Reflective Discussions (15 mins)

Reflections on the tool:

- Do you find the tool helpful to your work or research? If so, which part of the tool can best provide a benefit in your work?
- In terms of visualizing workflow and data manipulations, does the tool meet your expectations for clarity?
- Do you find the visual programming helpful in reducing your coding efforts?
- Regarding the analysis process, do you think the tool aids in streamlining your decision-making? Which specific feature of the tool do you find most supportive in this regard?
- At what points during your analysis do you find the visualization method recommendation helpful?
- Are you able to do something that you couldn't do before with this tool？

Usage scenario:

- Could you please describe how you will integrate this tool into your workflow? Are there specific tasks where the tool proves to be particularly beneficial?

Future directions:

- What additional functionalities or enhancements would you like to see in future versions of the tool to better accommodate your work or research needs?
- Are there areas where you think the tool could be refined or improved to better suit your professional requirements or to enhance user experience overall?

Part 3: Post-survey (5 mins)

Link to the survey:
https://qualtrics.kcl.ac.uk/jfe/form/SV_8HbjbP7olhMIqd8